

From Bellman Consistency toward Bellman Optimality: Constrained Online Critic Learning for Nonlinear Optimal Control

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Abstract—This paper proposes a constrained online critic-learning method for discrete-time control-affine nonlinear systems that moves beyond policy-dependent Bellman residual reduction toward Bellman optimality. Because a small policy-dependent residual does not verify minimization over alternative controls, the method formulates online critic learning as a constrained optimization problem, with the one-step Bellman backup as the objective and Bellman consistency as an equality constraint. The resulting Lagrangian-based law updates the critic weights and Bellman multiplier from sampled transitions and uses the critic’s state gradient for policy improvement. Lyapunov analysis establishes uniform ultimate boundedness of the learning errors. Real-world autonomous mobile robot experiments show that its trajectory root-mean-square error against an offline dynamic programming reference is 73.8% and 74.9% lower than those of one-step temporal-difference learning (TD(0)) and normalized adaptive dynamic programming, respectively. Complementary simulations show that its accumulated Bellman optimality residual over the evaluated state-error grid is 52.2% and 62.4% lower than those of the same methods, respectively. Additional learning reduces the residual along a newly encountered trajectory without substantially increasing it around the original trajectory.

Index Terms—Bellman optimality, constrained optimization, nonlinear optimal control, online critic learning.

I. INTRODUCTION

A. Background

Bellman’s principle of optimality provides the fundamental recursive structure underlying dynamic programming (DP) and optimal control [1]. It states that an optimal policy must remain optimal from any state reached after an initial decision. In discrete-time systems, this principle leads to the Bellman optimality equation, which characterizes the optimal value function through two coupled requirements. First, the one-step Bellman backup must be minimized with respect to the control input. Second, the minimized backup must be consistent with the value assigned to the current state. Together, these requirements connect value evaluation and optimal policy generation.

Exact DP solves the Bellman optimality equation by repeatedly evaluating the Bellman backup over the state–action space. Representative solution procedures include value iteration (VI), which directly applies the Bellman optimality operator, and policy iteration (PI), which alternates between policy evaluation and policy improvement. However, exact DP is generally intractable for continuous or high-dimensional dynamical systems because of the curse of dimensionality. Exact Bellman backups also typically require prior knowledge

of the system dynamics to determine the successor state corresponding to each candidate action.

To alleviate these limitations, adaptive dynamic programming (ADP) and reinforcement learning (RL) approximate value functions and policies using function approximation and sampled system transitions [2]–[4]. In the control literature, ADP broadly refers to approximate value function and policy learning for dynamical systems, which may be performed offline using collected data or online during system operation. Approximate VI seeks to reproduce the Bellman optimality backup, whereas approximate PI decomposes the problem into approximate policy evaluation and policy improvement. Actor–critic methods implement a related generalized policy-iteration structure by interleaving critic learning with policy improvement.

In these learning architectures, the critic approximates the value function and provides the information used to evaluate or improve the control policy. When critic adaptation is performed online, its parameters are updated incrementally using transitions generated during system operation. This capability is attractive for real-world control because it allows the critic and policy to improve while interacting with the physical system. The present study focuses on how such transition-based critic adaptation relates to the two structural requirements of the Bellman optimality equation.

B. Related Work

ADP has been extensively investigated for optimal control of continuous- and discrete-time nonlinear systems. VI-based studies have examined convergence of the approximate value sequence and admissibility of the resulting control policies, whereas PI-based and actor–critic methods have developed online policy-evaluation and improvement schemes with closed-loop stability guarantees [5]–[10]. Model-free and data-driven implementations have further reduced dependence on explicit system dynamics through integral formulations, system identification, and sampled transitions [11]–[14]. These studies establish the critic as a central component connecting value approximation and policy improvement in ADP-based nonlinear control.

Within ADP, RL, and actor–critic frameworks, temporal-difference (TD) and residual-gradient methods provide representative mechanisms for critic adaptation. TD(0) updates the critic using the error between the current value estimate and a one-step target generated from an observed transition. Because

the target is treated as fixed, TD(0) uses a semi-gradient update and is computationally efficient for online implementation. Residual-gradient methods instead differentiate the complete squared Bellman residual with respect to the critic parameters, accounting for the parameter dependence of both the current value estimate and the bootstrapped target [3]. This directly addresses residual minimization but may increase sensitivity to gradient scaling and numerical conditioning. In this paper, a normalized residual-gradient implementation adopted for comparison is referred to as normalized ADP [2], [8], [15].

Beyond these basic updates, online critic-learning methods have been extended using recursive least-squares adaptation, integral RL, identifier-assisted learning, concurrent learning, experience replay, and simulated transitions [11]–[13], [15]–[17]. Critic-only architectures and stability-guaranteed VI and PI schemes have also been developed to simplify policy implementation and analyze the interaction between learning and closed-loop dynamics. These variations aim to relax model and excitation requirements, improve data utilization, accelerate critic adaptation, or establish boundedness of the critic error and system state. Nevertheless, their critic-update signals are generally constructed from Bellman or Hamilton–Jacobi–Bellman (HJB) errors evaluated at observed or additionally generated transitions under a selected control policy.

Reducing such a sampled Bellman residual, however, does not necessarily translate directly into an accurate value function or an improved control policy. In online learning, both the sampled states and the evaluated actions depend on the behavior of the current closed-loop policy. The resulting empirical objective is therefore weighted by a sampling distribution determined by the policy, initial conditions, disturbances, and exploration. A small empirical residual under this distribution may coexist with substantial value errors because of error cancellation and incomplete state-action coverage [18]. Moreover, analyses of continuous-state control problems have identified non-stabilizing solutions of Bellman and HJB equations, highlighting the importance of additional conditions on the learned value function [19]. Bellman-residual magnitude alone is therefore not a sufficient indicator of critic quality or control performance.

Recent studies have distinguished Bellman-residual minimization based on a selected policy from that based on the Bellman optimality operator [4], [20]. In the terminology of this paper, the corresponding residuals are referred to as the policy-dependent Bellman residual and the Bellman optimality residual, respectively. The former measures value consistency using the action prescribed by the current policy, whereas the latter measures consistency after the one-step Bellman backup is minimized over the control input. These residuals coincide only when the critic accurately evaluates a policy that is also greedy with respect to the resulting value function, which is generally not the case during online approximate learning.

Complementary approaches have pursued Bellman optimality through either the structure of the critic or the policy-improvement mechanism. The positive-definite value-function architecture in [19] excludes non-stabilizing solutions by re-

stricting the critic to positive-definite candidates. Lin et al. [21] integrate RL and model predictive control (MPC) through policy iteration: RL evaluates the current policy, while the learned value function is used as the terminal cost of an MPC problem that generates an improved policy. Thus, control-input minimization is performed explicitly by MPC, whereas Bellman optimality is approached indirectly through repeated policy evaluation and MPC-based policy improvement rather than enforced directly in the critic update. These studies pursue Bellman optimality through critic structure and MPC-based policy improvement, respectively. However, one-step action minimization and value consistency remain handled separately rather than being directly coupled within sample-based online critic adaptation. This leaves open how both elements of Bellman optimality can be incorporated into a critic update using streaming system transitions.

C. Contributions

To address this gap, this paper develops a constrained online critic-learning method for discrete-time control-affine nonlinear systems. Related ADP studies have employed constrained formulations primarily to account for requirements imposed on the underlying optimal control problem. For example, Bandyopadhyay and Bhasin [22] use a Lagrangian formulation to handle state constraints in online critic learning, whereas Wei et al. [23] employ a primal–dual formulation for isoperimetric constraints in finite-horizon optimal control. In contrast, the proposed formulation treats one-step Bellman-backup minimization as the optimization objective and imposes consistency between the minimized backup and the current critic estimate as an equality constraint. This formulation places action minimization and value consistency within a common sample-based online learning structure without requiring exhaustive Bellman backups over the entire state–action space. An overview of the proposed framework is provided in Fig. 1.

The main contributions of this paper are summarized as follows:

- A constrained optimization formulation is established that uses the one-step Bellman backup as the objective and imposes value consistency as an equality constraint, thereby coupling the two elements of Bellman optimality within sample-based online critic learning.
- A Lagrangian-based online learning law is derived to jointly update the critic weights and the associated Lagrange multiplier, termed the Bellman multiplier, from sampled system transitions. An online policy-improvement law based on the learned critic gradient is also developed for control-affine nonlinear systems.
- A Lyapunov analysis establishes uniform ultimate boundedness of the critic-weight and Bellman-multiplier errors under the stated boundedness and function-approximation conditions.
- Real-world autonomous mobile robot (AMR) experiments and complementary simulations compare the proposed method with an offline DP reference, TD(0),

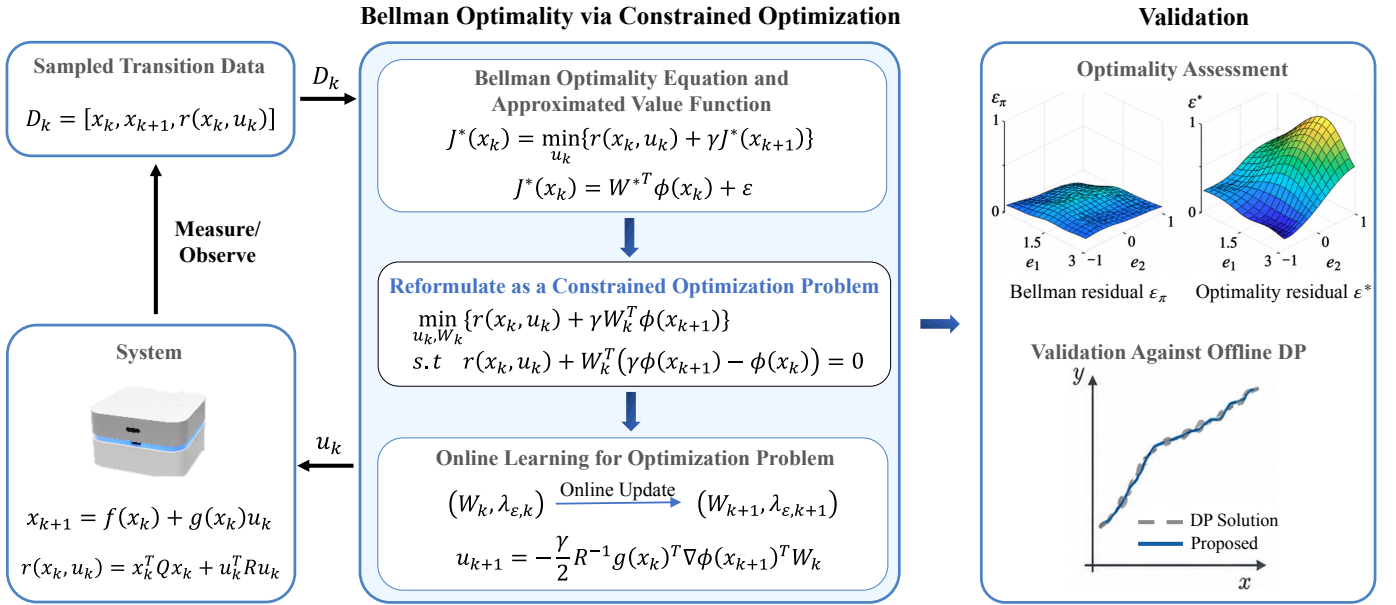


Fig. 1: Contributions diagram.

and normalized ADP. Residual-surface and critic-freeze analyses demonstrate closer agreement with the offline DP reference in the examined regions, while additional learning improves Bellman optimality consistency in a newly visited region without substantially degrading it near the original trajectory.

D. Organization

The remainder of this paper is organized as follows. Section II formulates the nonlinear optimal control problem and introduces the policy-dependent Bellman residual and Bellman optimality residual. Section III develops the proposed constrained online learning method. Section IV presents the learning-stability analysis. Section V reports the real-world AMR experiments and simulation-based analyses, and Section VI concludes the paper.

II. PRELIMINARIES

A. Nonlinear Optimal Control Problem

Consider a class of discrete-time, control-affine nonlinear systems described by

$$x_{k+1} = f(x_k) + g(x_k)u_k, \quad (1)$$

where $k \in \mathbb{Z}_{\geq 0}$ is the discrete-time index, $x_k \in \mathbb{R}^n$ is the state vector, and $u_k \in \mathbb{R}^m$ is the control input. The function $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ represents the drift dynamics, and $g: \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ denotes the control input effectiveness.

The objective is to determine a state-feedback policy $u_k = \pi(x_k)$ that minimizes the infinite-horizon discounted cost

$$J(x_k) = \sum_{i=k}^{\infty} \gamma^{i-k} r(x_i, u_i), \quad (2)$$

where $r(x_i, u_i)$ is the stage cost and $\gamma \in (0, 1]$ is the discount factor. In this work, the stage cost is considered in the form

$$r(x_k, u_k) = Q(x_k) + u_k^T R u_k, \quad (3)$$

where $Q(x_k) \geq 0$ is the state cost and $R \succ 0$ is the input weighting matrix.

According to Bellman's principle of optimality, the optimal value function $J^*(x_k)$ satisfies the Bellman optimality equation:

$$J^*(x_k) = \min_{u_k} \{r(x_k, u_k) + \gamma J^*(x_{k+1})\}. \quad (4)$$

The corresponding optimal policy is obtained by

$$\pi^*(x_k) = \arg \min_{u_k} \{r(x_k, u_k) + \gamma J^*(x_{k+1})\}. \quad (5)$$

For the control-affine dynamics in (1) and the input-quadratic stage cost in (3), a local discrete-time policy improvement form can be written as

$$u_k^* = -\frac{\gamma}{2} R^{-1} g(x_k)^T \nabla_x J^*(x_{k+1}), \quad (6)$$

where $\nabla_x J^*$ denotes the state gradient of the value function.

B. Bellman Residual and Bellman Optimality Residual

For a candidate value function $J: \Omega \rightarrow \mathbb{R}$, the Bellman residual is evaluated using the sampled transition data $(x_k, x_{k+1}, r(x_k, u_k))$ as

$$\varepsilon_\pi(x_k) = r(x_k, u_k) + \gamma J(x_{k+1}) - J(x_k). \quad (7)$$

Here, u_k is the input generated by the currently applied policy and x_{k+1} is the corresponding successor state. Thus, $\varepsilon_\pi(x_k)$ evaluates the policy-dependent consistency of the candidate value function along the sampled transition.

In contrast, the Bellman optimality residual evaluates consistency with the minimization structure of the Bellman optimality equation:

$$\varepsilon^*(x_k) = \min_{u \in \mathcal{U}} \{r(x_k, u) + \gamma J(f(x_k) + g(x_k)u)\} - J(x_k). \quad (8)$$

Unlike $\varepsilon_\pi(x_k)$, which evaluates the Bellman relation at the input generated by the currently applied policy, $\varepsilon^*(x_k)$ explicitly includes the minimization required by the Bellman optimality equation. Therefore, a small Bellman residual indicates policy-dependent consistency along the sampled transition, whereas a small Bellman optimality residual indicates consistency with the optimal Bellman backup.

III. PROPOSED METHOD

The proposed method is developed to improve the consistency of online critic learning with the Bellman optimality equation. Instead of updating the critic only by reducing the sampled Bellman residual, the proposed method reformulates the one-step Bellman optimality relation as a constrained optimization problem. The Bellman optimality backup is used as the learning objective, while the Bellman equation is imposed as an equality constraint. This formulation leads to a Lagrangian-based online update law for critic learning.

A. Function Approximation

Since an analytical expression for the optimal value function $J^*(x_k)$ is generally unavailable for nonlinear systems, a function approximation scheme is introduced. Assuming that $J^*(x_k)$ is continuous over a compact set $\Omega \subset \mathbb{R}^n$, the Weierstrass approximation theorem implies that $J^*(x_k)$ can be approximated with arbitrary accuracy by a sufficiently rich set of basis functions. Accordingly, the optimal value function is approximated as

$$J^*(x_k) = W^{*T} \phi(x_k) + \varepsilon(x_k), \quad (9)$$

where $\phi(x_k) \in \mathbb{R}^p$ is the basis function vector, $W^* \in \mathbb{R}^p$ is the ideal critic weight vector, and $\varepsilon(x_k)$ is the approximation error. The ideal critic weight is defined as

$$W^* = \arg \min_{W \in \mathbb{R}^p} \left\{ \sup_{x_k \in \Omega} \|J^*(x_k) - W^T \phi(x_k)\| \right\}. \quad (10)$$

In online learning, the critic approximation is given by

$$\hat{J}(x_k, W_k) = W_k^T \phi(x_k). \quad (11)$$

B. Constrained Optimization Formulation for Bellman Optimality

Substituting the critic approximation in (11) into the Bellman optimality equation in (4) yields the approximate Bellman optimality equation

$$W_k^T \phi(x_k) = \min_{u_k} \{r(x_k, u_k) + \gamma W_k^T \phi(x_{k+1})\}. \quad (12)$$

For a given state transition, the temporal basis difference is defined as

$$\zeta_k = \gamma \phi(x_{k+1}) - \phi(x_k), \quad (13)$$

and the corresponding Bellman residual is expressed as

$$\begin{aligned} \varepsilon_k &= r(x_k, u_k) + \gamma W_k^T \phi(x_{k+1}) - W_k^T \phi(x_k) \\ &= r(x_k, u_k) + W_k^T \zeta_k. \end{aligned} \quad (14)$$

Here, ε_k denotes the policy-dependent Bellman residual $\varepsilon_\pi(x_k)$ evaluated using the current critic weights W_k and the sampled transition.

The approximate Bellman optimality equation in (12) contains two coupled requirements. First, the one-step Bellman backup must be minimized with respect to the control input. Second, the minimized backup must be consistent with the current value estimate. The Bellman residual condition $\varepsilon_k = 0$ represents only the latter requirement for a given control input and, by itself, does not determine whether the input minimizes the Bellman backup.

To incorporate both requirements, the control input and critic weight are treated as primal decision variables. The proposed constrained optimization problem for Bellman optimality is formulated as

$$\begin{aligned} \min_{u_k, W_k} \quad & r(x_k, u_k) + \gamma W_k^T \phi(x_{k+1}) \\ \text{subject to} \quad & r(x_k, u_k) + \gamma W_k^T \phi(x_{k+1}) - W_k^T \phi(x_k) = 0. \end{aligned} \quad (15)$$

Equivalently, the equality constraint in (15) is expressed as $\varepsilon_k(u_k, W_k) = 0$. The objective retains the minimization structure of the Bellman optimality equation, while the equality constraint enforces consistency between the current value estimate and the one-step Bellman backup. Thus, the formulation explicitly distinguishes Bellman residual satisfaction from Bellman optimality.

The Lagrangian associated with (15) is defined as

$$\begin{aligned} L_k(u_k, W_k, \lambda_{\varepsilon, k}) &= r(x_k, u_k) + \gamma W_k^T \phi(x_{k+1}) \\ &\quad + \lambda_{\varepsilon, k} [r(x_k, u_k) + W_k^T \zeta_k], \end{aligned} \quad (16)$$

where $\lambda_{\varepsilon, k}$ is the Lagrange multiplier associated with the Bellman equality constraint.

Under the standard regularity condition for the equality constraint, a local solution of (15) satisfies the following Karush–Kuhn–Tucker (KKT) conditions:

$$\nabla_{u_k} L_k = 0, \quad (17)$$

$$\nabla_{W_k} L_k = 0, \quad (18)$$

$$\frac{\partial L_k}{\partial \lambda_{\varepsilon, k}} = \varepsilon_k(u_k, W_k) = 0. \quad (19)$$

Equation (17) represents the stationarity condition associated with policy improvement, whereas (18) represents the stationarity condition for the critic weight. Equation (19) enforces the Bellman equality condition through the multiplier.

The KKT conditions characterize the ideal sample-wise solution of the constrained problem. In the proposed online implementation, this solution is not computed to convergence at every sampling instant. Instead, the critic weight and Bellman multiplier are incrementally updated using each sampled transition, followed by a policy improvement step.

C. Online Learning Law

The proposed online learning law follows the basic structure of discrete-time policy iteration. At each sampling instant, the current input u_k is applied to the system, and the transition data $(x_k, x_{k+1}, r(x_k, u_k))$ are obtained. The critic weight and Bellman multiplier are then updated using the sampled transition in the policy evaluation step, after which the next control input is computed through policy improvement.

The policy evaluation step is formulated using the proposed constrained learning structure. Inspired by the augmented Lagrangian method, the Bellman equality residual is incorporated through a multiplier, and a residual-dependent correction term is added to improve online learning stability.

1) *Critic Weight Update:* In the policy evaluation step, the sampled transition is used to update the critic weights as

$$W_{k+1} = W_k - \alpha_W (\nabla_W L_k + \mu \zeta_k \varepsilon_k + \sigma_W W_k), \quad (20)$$

where $\alpha_W > 0$ is the critic learning rate, $\mu > 0$ is the residual-penalty coefficient, and $\sigma_W > 0$ is a small coefficient introduced for learning stability. In (20), $\nabla_W L_k$ is the critic-stationarity residual appearing in KKT condition (18), while $\mu \zeta_k \varepsilon_k$ is the correction associated with the quadratic residual penalty $\frac{\mu}{2} \varepsilon_k^2$ and reinforces the Bellman equality condition in (19).

From (16), the gradient with respect to the critic weight is obtained as

$$\begin{aligned} \nabla_W L_k &= \frac{\partial L_k}{\partial W_k} \\ &= \gamma \phi(x_{k+1}) + \lambda_{\varepsilon,k} (\gamma \phi(x_{k+1}) - \phi(x_k)) \\ &= \gamma \phi(x_{k+1}) + \lambda_{\varepsilon,k} \zeta_k. \end{aligned} \quad (21)$$

Therefore, the expanded critic update law becomes

$$W_{k+1} = W_k - \alpha_W (\gamma \phi(x_{k+1}) + \lambda_{\varepsilon,k} \zeta_k + \mu \zeta_k \varepsilon_k + \sigma_W W_k). \quad (22)$$

2) *Multiplier Update:* In the same policy evaluation step, the Bellman multiplier is updated by

$$\lambda_{\varepsilon,k+1} = \lambda_{\varepsilon,k} + \mu (\varepsilon_k - \sigma_\lambda \lambda_{\varepsilon,k}), \quad (23)$$

where $\sigma_\lambda > 0$ is a small stability coefficient. Since $\partial L_k / \partial \lambda_{\varepsilon,k} = \varepsilon_k$, the term $\mu \varepsilon_k$ implements dual ascent using the feasibility residual in KKT condition (19), while $-\mu \sigma_\lambda \lambda_{\varepsilon,k}$ regularizes the multiplier update to support learning stability.

3) *Policy Improvement:* After the sampled transition is used for policy evaluation, the control input for the next sampling instant is computed from the local discrete-time policy improvement structure:

$$u_{k+1} = -\frac{\gamma}{2} R^{-1} g(x_k)^T \nabla \phi(x_{k+1})^T W_k. \quad (24)$$

Here, u_{k+1} denotes the input to be applied at the next sampling instant, and W_k is used because the policy-improvement law is obtained from the control-input stationarity condition in (17) using the current Bellman evaluation.

Algorithm 1 summarizes the proposed constrained online critic learning procedure.

Algorithm 1 Proposed Constrained Online Critic Learning toward Bellman Optimality

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1: Initialize critic weights  $W_0$  and Bellman multiplier  $\lambda_{\varepsilon,0}$ 
2: Set  $\alpha_W, \mu, \sigma_W, \sigma_\lambda, \gamma, Q$ , and  $R$ 
3: Measure initial state  $x_0$  and set initial input  $u_0 = 0$ 
4: for  $k = 0, 1, 2, \dots$  do
5:   Apply current policy and collect transition data:
6:   Apply  $u_k$  to the system and measure  $x_{k+1}$ 
7:   Compute the stage cost  $r(x_k, u_k)$ 
8:   Bellman residual evaluation:
9:    $\zeta_k \leftarrow \gamma \phi(x_{k+1}) - \phi(x_k)$ 
10:   $\varepsilon_k \leftarrow r(x_k, u_k) + W_k^T \zeta_k$ 
11:  Policy evaluation:
12:   $\nabla_{W_k} L_k \leftarrow \gamma \phi(x_{k+1}) + \lambda_{\varepsilon,k} \zeta_k$ 
13:   $W_{k+1} \leftarrow W_k - \alpha_W (\nabla_{W_k} L_k + \mu \zeta_k \varepsilon_k + \sigma_W W_k)$ 
14:   $\lambda_{\varepsilon,k+1} \leftarrow \lambda_{\varepsilon,k} + \mu (\varepsilon_k - \sigma_\lambda \lambda_{\varepsilon,k})$ 
15:  Policy improvement:
16:   $u_{k+1} \leftarrow -\frac{\gamma}{2} R^{-1} g(x_k)^T \nabla \phi(x_{k+1})^T W_k$ 
17:  Time update:
18:   $x_k \leftarrow x_{k+1}, u_k \leftarrow u_{k+1}$ 
19:   $W_k \leftarrow W_{k+1}, \lambda_{\varepsilon,k} \leftarrow \lambda_{\varepsilon,k+1}$ 
20: end for

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IV. LEARNING STABILITY ANALYSIS

This section analyzes the learning stability of the proposed critic-learning method using Lyapunov theory. The learning error relations are first established using perturbed stationarity and Bellman equality conditions at an optimal solution pair. An upper bound on the discrete-time Lyapunov difference is then derived. Under the stated residual and update-increment boundedness assumptions, the critic-weight and Bellman-multiplier errors are shown to be uniformly ultimately bounded.

Let W^* and λ_ε^* denote the ideal critic weight and Bellman multiplier satisfying the KKT conditions of the constrained optimization problem. Due to function approximation errors, the stationarity and primal feasibility conditions are written as

$$\gamma \phi(x_{k+1}) + \lambda_\varepsilon^* \zeta_k + \rho_{s,k} = 0, \quad (25)$$

$$r(x_k, u_k) + (W^*)^T \zeta_k + \rho_{r,k} = 0, \quad (26)$$

where $\rho_{s,k}$ and $\rho_{r,k}$ represent the mismatch in the stationarity and Bellman equality relations, respectively, including the effects of function approximation. These perturbed relations are used below to describe the reference pair at each evaluated transition. The residual terms are taken to satisfy the uniform bounds

$$\|\rho_{s,k}\| \leq \bar{\rho}_s, \quad |\rho_{r,k}| \leq \bar{\rho}_r, \quad (27)$$

where the bounding constants are finite and independent of the initial time.

Define the learning errors relative to the reference pair $(W^*, \lambda_\varepsilon^*)$ as

$$\tilde{W}_k = W_k - W^*, \quad \tilde{\lambda}_{\varepsilon,k} = \lambda_{\varepsilon,k} - \lambda_\varepsilon^*. \quad (28)$$

Using (21) and (25), the gradient evaluated at $(W_k, \lambda_{\varepsilon,k})$ can be expressed in terms of the multiplier error as

$$\begin{aligned}\nabla_W L_k &= \gamma \phi(x_{k+1}) + \lambda_{\varepsilon,k} \zeta_k \\ &= (\gamma \phi(x_{k+1}) + \lambda_{\varepsilon}^* \zeta_k) + \zeta_k (\lambda_{\varepsilon,k} - \lambda_{\varepsilon}^*) \\ &= \zeta_k \tilde{\lambda}_{\varepsilon,k} - \rho_{s,k}.\end{aligned}\quad (29)$$

Similarly, (14) and (26) give

$$\begin{aligned}\varepsilon_k &= r(x_k, u_k) + W_k^T \zeta_k \\ &= (r(x_k, u_k) + (W^*)^T \zeta_k) + \zeta_k^T (W_k - W^*) \\ &= \zeta_k^T \tilde{W}_k - \rho_{r,k}.\end{aligned}\quad (30)$$

Let $\alpha_W, \mu, \sigma_W, \sigma_\lambda > 0$ be constant. Based on the learning errors defined above, select the discrete-time Lyapunov function candidate

$$V_k = \frac{1}{2\alpha_W} \|\tilde{W}_k\|^2 + \frac{1}{2\mu} \tilde{\lambda}_{\varepsilon,k}^2. \quad (31)$$

This function is positive definite and radially unbounded with respect to the learning error vector.

Assumption 1: Define the learning increments as

$$\Delta W_k = W_{k+1} - W_k, \quad \Delta \lambda_{\varepsilon,k} = \lambda_{\varepsilon,k+1} - \lambda_{\varepsilon,k}. \quad (32)$$

For each prescribed bounded set of initial learning errors, these increments satisfy

$$\|\Delta W_k\|^2 \leq \overline{\Delta W}^2, \quad (\Delta \lambda_{\varepsilon,k})^2 \leq \overline{\Delta \lambda}^2, \quad (33)$$

uniformly over the considered trajectories and initial times, where the bounding constants are finite. The magnitudes of the learning increments can be regulated through the adjustable learning gains α_W and μ in the update laws. $\diamond$

Theorem 1: Suppose that the residual bounds in (27) and Assumption 1 hold. For positive gains $\alpha_W, \mu, \sigma_W, \sigma_\lambda$ satisfying $0 < \alpha_V < 1$, where

$$\alpha_V = \min \left\{ \frac{\alpha_W \sigma_W}{2}, \frac{\mu \sigma_\lambda}{2} \right\}, \quad (34)$$

the critic-weight and Bellman-multiplier errors relative to the reference pair $(W^*, \lambda_{\varepsilon}^*)$ associated with the perturbed KKT conditions are uniformly ultimately bounded. $\diamond$

Proof: Using (28) and (32), the learning errors satisfy

$$\begin{aligned}\tilde{W}_{k+1} &= \tilde{W}_k + \Delta W_k, \\ \tilde{\lambda}_{\varepsilon,k+1} &= \tilde{\lambda}_{\varepsilon,k} + \Delta \lambda_{\varepsilon,k}.\end{aligned}\quad (35)$$

Using (31), the exact difference $\Delta V_k = V_{k+1} - V_k$ is

$$\begin{aligned}\Delta V_k &= \frac{1}{\alpha_W} \tilde{W}_k^T \Delta W_k + \frac{1}{2\alpha_W} \|\Delta W_k\|^2 \\ &\quad + \frac{1}{\mu} \tilde{\lambda}_{\varepsilon,k} \Delta \lambda_{\varepsilon,k} + \frac{1}{2\mu} (\Delta \lambda_{\varepsilon,k})^2.\end{aligned}\quad (36)$$

By Assumption 1, the second-order increment terms satisfy

$$\frac{1}{2\alpha_W} \|\Delta W_k\|^2 + \frac{1}{2\mu} (\Delta \lambda_{\varepsilon,k})^2 \leq \Gamma_I, \quad (37)$$

where

$$\Gamma_I = \frac{\overline{\Delta W}^2}{2\alpha_W} + \frac{\overline{\Delta \lambda}^2}{2\mu}. \quad (38)$$

Substituting the update laws in (20) and (23) into (36) and applying (37) yield

$$\begin{aligned}\Delta V_k &\leq -\tilde{W}_k^T (\nabla_W L_k + \mu \zeta_k \varepsilon_k + \sigma_W W_k) \\ &\quad + \tilde{\lambda}_{\varepsilon,k} (\varepsilon_k - \sigma_\lambda \lambda_{\varepsilon,k}) + \Gamma_I.\end{aligned}\quad (39)$$

Using (29) and (30), the cross terms $-\tilde{W}_k^T \zeta_k \tilde{\lambda}_{\varepsilon,k}$ and $\tilde{\lambda}_{\varepsilon,k} \zeta_k^T \tilde{W}_k$ cancel. Consequently,

$$\begin{aligned}\Delta V_k &\leq -\mu \tilde{W}_k^T \zeta_k \varepsilon_k + \tilde{W}_k^T \rho_{s,k} - \tilde{\lambda}_{\varepsilon,k} \rho_{r,k} \\ &\quad - \sigma_W \tilde{W}_k^T W_k - \sigma_\lambda \tilde{\lambda}_{\varepsilon,k} \lambda_{\varepsilon,k} + \Gamma_I.\end{aligned}\quad (40)$$

Since $\tilde{W}_k^T \zeta_k = \varepsilon_k + \rho_{r,k}$, this becomes

$$\begin{aligned}\Delta V_k &\leq -\mu \varepsilon_k^2 - \mu \rho_{r,k} \varepsilon_k + \tilde{W}_k^T \rho_{s,k} - \tilde{\lambda}_{\varepsilon,k} \rho_{r,k} \\ &\quad - \sigma_W \tilde{W}_k^T W_k - \sigma_\lambda \tilde{\lambda}_{\varepsilon,k} \lambda_{\varepsilon,k} + \Gamma_I.\end{aligned}\quad (41)$$

Using (27), Young's inequality gives

$$-\mu \rho_{r,k} \varepsilon_k \leq \frac{\mu}{4} \varepsilon_k^2 + \mu \bar{\rho}_r^2, \quad (42)$$

$$\tilde{W}_k^T \rho_{s,k} \leq \frac{\sigma_W}{4} \|\tilde{W}_k\|^2 + \frac{\bar{\rho}_s^2}{\sigma_W}, \quad (43)$$

$$-\tilde{\lambda}_{\varepsilon,k} \rho_{r,k} \leq \frac{\sigma_\lambda}{4} \tilde{\lambda}_{\varepsilon,k}^2 + \frac{\bar{\rho}_r^2}{\sigma_\lambda}. \quad (44)$$

Furthermore, since $W_k = \tilde{W}_k + W^*$ and $\lambda_{\varepsilon,k} = \tilde{\lambda}_{\varepsilon,k} + \lambda_{\varepsilon}^*$, the terms involving σ_W and σ_λ satisfy

$$-\sigma_W \tilde{W}_k^T W_k \leq -\frac{\sigma_W}{2} \|\tilde{W}_k\|^2 + \frac{\sigma_W}{2} \|W^*\|^2, \quad (45)$$

and

$$\begin{aligned}-\sigma_\lambda \tilde{\lambda}_{\varepsilon,k} \lambda_{\varepsilon,k} &\leq -\frac{\sigma_\lambda}{2} \tilde{\lambda}_{\varepsilon,k}^2 \\ &\quad + \frac{\sigma_\lambda}{2} (\lambda_{\varepsilon}^*)^2.\end{aligned}\quad (46)$$

Substituting (42)–(46) into (41) yields

$$\begin{aligned}\Delta V_k &\leq -\frac{3}{4} \mu \varepsilon_k^2 - \frac{\sigma_W}{4} \|\tilde{W}_k\|^2 \\ &\quad - \frac{\sigma_\lambda}{4} \tilde{\lambda}_{\varepsilon,k}^2 + \Gamma,\end{aligned}\quad (47)$$

where

$$\begin{aligned}\Gamma &= \Gamma_I + \mu \bar{\rho}_r^2 + \frac{\bar{\rho}_s^2}{\sigma_W} + \frac{\bar{\rho}_r^2}{\sigma_\lambda} \\ &\quad + \frac{\sigma_W}{2} \|W^*\|^2 + \frac{\sigma_\lambda}{2} (\lambda_{\varepsilon}^*)^2.\end{aligned}\quad (48)$$

By (31) and (34),

$$\begin{aligned}\alpha_V V_k &= \frac{\alpha_V}{2\alpha_W} \|\tilde{W}_k\|^2 + \frac{\alpha_V}{2\mu} \tilde{\lambda}_{\varepsilon,k}^2 \\ &\leq \frac{\sigma_W}{4} \|\tilde{W}_k\|^2 + \frac{\sigma_\lambda}{4} \tilde{\lambda}_{\varepsilon,k}^2.\end{aligned}\quad (49)$$

Dropping the nonpositive residual term in (47) therefore gives

$$\Delta V_k \leq -\alpha_V V_k + \Gamma. \quad (50)$$

Since $0 < \alpha_V < 1$, the corresponding recursion $V_{k+1} \leq (1 - \alpha_V) V_k + \Gamma$ ensures that, for any $\eta > 0$, V_k enters and remains below $\Gamma/\alpha_V + \eta$ in finite time, uniformly in the initial time for bounded initial learning errors. The positive-definite quadratic form of (31) then establishes uniform ultimate boundedness of the learning errors. $\blacksquare$

V. VALIDATION

This section validates and analyzes the proposed constrained online critic-learning method through real-world AMR experiments and simulation studies. First, the target-reaching task, critic structure, comparative methods, and offline DP reference are introduced. Then, experimental results are presented to evaluate learning behavior across repeated episodes and compare the learned policies with the DP reference. Simulation-based analyses further examine gain sensitivity, Bellman optimality residuals, critic-freeze behavior, and additional learning from an unvisited trajectory region. An overview of the validation structure, including the roles of the experimental and simulation-based analyses, is provided in Fig. 2.

A. Problem Setting

1) *AMR Kinematics and Error Dynamics*: The validation problem is formulated as a target-reaching task of an AMR. The motion of the AMR is modeled by the following discrete-time kinematics:

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \\ \theta_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \\ \theta_k \end{bmatrix} + T_s \begin{bmatrix} v_k \cos \theta_k \\ v_k \sin \theta_k \\ \omega_k \end{bmatrix}, \quad (51)$$

where (x_k, y_k) denotes the position of the AMR, θ_k denotes its heading angle, and $u_k = [v_k, \omega_k]^T$ is the control input consisting of the linear and angular velocities.

The AMR starts from the initial pose

$$[x_0, y_0, \theta_0]^T = [0, 0, 0]^T, \quad (52)$$

and is required to reach the target position

$$[x_r, y_r]^T = [2, -1]^T. \quad (53)$$

The error is represented by

$$e_k = \begin{bmatrix} e_{1,k} \\ e_{2,k} \end{bmatrix}, \quad (54)$$

where the individual error components are defined as

$$\begin{aligned} e_{1,k} &= \sqrt{(x_r - x_k)^2 + (y_r - y_k)^2}, \\ e_{2,k} &= \text{atan2}(y_r - y_k, x_r - x_k) - \theta_k. \end{aligned} \quad (55)$$

Here, $e_{1,k}$ denotes the Euclidean distance from the AMR to the target, and $e_{2,k}$ denotes the heading error between the AMR heading and the line-of-sight direction to the target. In implementation, $e_{2,k}$ is wrapped to the interval $[-\pi, \pi)$ to maintain a consistent angular representation.

Using the same forward Euler approximation as in (51), the discrete-time error dynamics used for online learning are expressed as

$$e_{k+1} = e_k + T_s \begin{bmatrix} -\cos e_{2,k} & 0 \\ \frac{\sin e_{2,k}}{e_{1,k}} & -1 \end{bmatrix} u_k. \quad (56)$$

To avoid the coordinate singularity at the target, $e_{1,k}$ in the denominator is bounded below by a small positive constant in implementation.

2) *Cost Function, Critic Basis, and Offline DP Reference*: The stage cost is defined as

$$r(e_k, u_k) = e_k^T Q e_k + u_k^T R u_k, \quad (57)$$

with

$$Q = \text{diag}(10, 10), \quad R = \text{diag}(4.5, 4.5), \quad (58)$$

and the discount factor is set to $\gamma = 0.95$. The sampling time is $T_s = 0.01$ s.

The critic basis functions are selected as

$$\phi(e_k) = [e_{1,k}^2 \quad e_{2,k}^2 \quad e_{1,k}e_{2,k} \quad \cos e_{2,k} - 1 \quad e_{1,k} \sin e_{2,k}]^T. \quad (59)$$

This basis includes the individual error terms and nonlinear coupling terms that reflect the geometry of the AMR target-reaching problem.

The offline DP reference is obtained by solving the discretized Bellman optimality equation using VI. The error state space is discretized over (e_1, e_2) , and the input space is discretized over (v, ω) . At each error-state grid point, an exhaustive search over the discretized input space is performed using the Bellman backup

$$J_{i+1}(e) = \min_u \{r(e, u) + \gamma J_i(e^+)\}, \quad (60)$$

where i denotes the value-iteration index and e^+ denotes the one-step successor error state obtained from (56) for a given state-input pair (e, u) . The resulting offline DP trajectory and control inputs are used as numerical references for the trajectory and control-input root-mean-square error (RMSE) comparisons.

3) *Comparison Methods*: Three online critic-learning methods are compared under the same task, cost function, sampling time, critic basis, and policy structure. The proposed method is the constrained online critic-learning method toward Bellman optimality presented in Algorithm 1. For comparison, TD(0) learning [3] and normalized ADP [15] are selected as representative TD-error-based online critic-learning methods.

TD(0) learning is used as a standard one-step temporal-difference comparative method. It updates the critic directly from the sampled Bellman residual as

$$W_{k+1} = W_k + \alpha_W \epsilon_k \phi(e_k). \quad (61)$$

The normalized ADP method is implemented as a normalized residual-gradient ADP method. This method updates the critic in the direction of the Bellman residual gradient while using a squared normalization term to reduce sensitivity to large temporal basis differences:

$$W_{k+1} = W_k - \alpha_W \frac{\zeta_k \epsilon_k}{(1 + \zeta_k^T \zeta_k)^2}. \quad (62)$$

All methods use the same initial critic weights. Since the proposed method, TD(0) learning, and normalized ADP have different critic update structures, their learning gains cannot be fairly compared by assigning the same numerical value. Therefore, the nominal learning gains for each method were

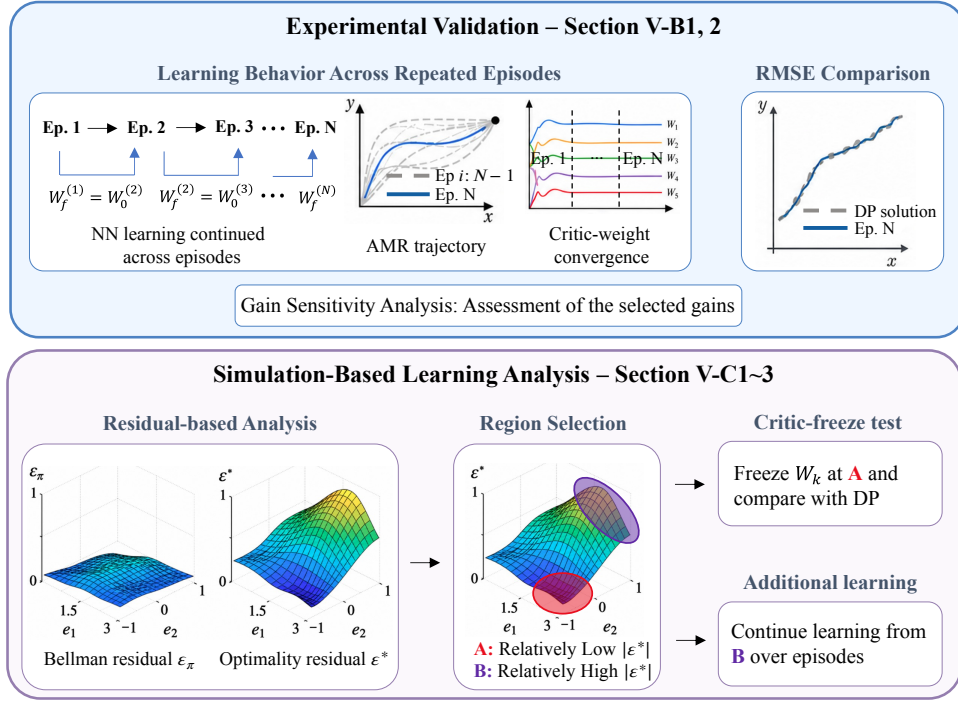


Fig. 2: Overview of the validation structure.

selected separately through repeated empirical tuning in real-world AMR experiments. The selected gains correspond to the best observed experimental performance obtained for each method under the same task setting.

Specifically, the proposed method used $\alpha_W = 1 \times 10^{-3}$ and $\mu = 0.017$, with stability coefficients $\sigma_W = \sigma_\lambda = 10^{-6}$. TD(0) learning used $\alpha_W = 3 \times 10^{-3}$, and normalized ADP used $\alpha_W = 3 \times 10^{-2}$. To further support this gain selection procedure, a simulation-based gain sensitivity analysis is presented in Section V-B2 after the experimental comparison. This analysis confirms that the experimental gains lie within reasonable operating regions for their respective methods.

Table I summarizes the common validation parameters.

B. Experimental Results

The proposed method and the comparative methods were implemented on the AMR platform shown in Fig. 3. The results in this subsection are organized into two parts. First, results across repeated episodes are presented to evaluate how each method adapts the critic online and improves closed-loop trajectory performance as the same target-reaching task is

repeated. Second, a simulation-based gain sensitivity analysis is presented to assess the reasonableness of the learning gains used in the AMR experiments.

1) *Learning Behavior Across Repeated Episodes*: Each method was evaluated over multiple episodes of the same target-reaching task. During each episode, the critic weights were updated online at every sampling instant. The weights obtained at the end of each episode were retained and used to initialize the subsequent episode. This protocol is used to examine whether online adaptation leads to progressively improved and repeatable closed-loop behavior across episodes.

Figures 4–6 show the learning behavior of the proposed and comparative methods across repeated episodes. For each episode, the figures present the AMR trajectory together with the corresponding critic-weight evolution. Each panel therefore illustrates both the closed-loop AMR motion and online critic adaptation during the same task execution.

As shown in Fig. 4, the proposed method rapidly reshapes the initial trajectory during the first episode and produces nearly repeatable trajectories in the subsequent episodes. The

TABLE I: Validation Parameters

Parameter	Value
Initial pose	$[0, 0, 0]^T$
Target position	$[2, -1]^T$
Sampling time T_s	10 ms
Discount factor γ	0.95
State weighting Q	$\text{diag}(10, 10)$
Input weighting R	$\text{diag}(4.5, 4.5)$
Initial critic weight W_0	$[30, 30, 30, 30, 30]^T$



Fig. 3: AMR platform used in the experimental validation.

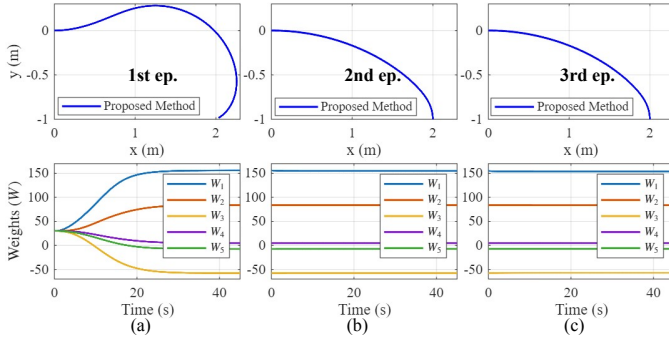


Fig. 4: Learning behavior of the proposed method across repeated episodes: (a) 1st, (b) 2nd, and (c) 3rd episodes. Each panel shows the AMR trajectory and critic-weight evolution during the corresponding episode.

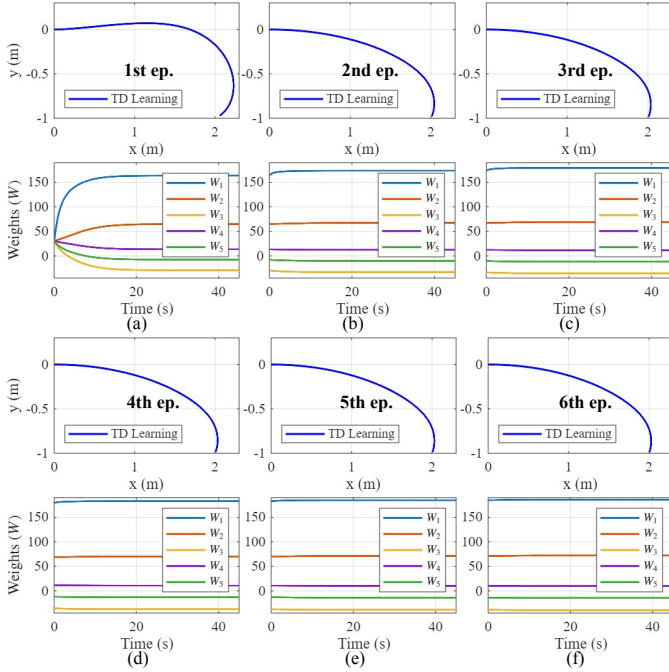


Fig. 5: Learning behavior of TD(0) across repeated episodes: (a) 1st, (b) 2nd, (c) 3rd, (d) 4th, (e) 5th, and (f) 6th episodes. Each panel shows the AMR trajectory and critic-weight evolution during the corresponding episode.

critic-weight evolution also exhibits only minor changes after the first task execution. This indicates that the information acquired during the initial execution is retained and supports repeatable closed-loop behavior in subsequent episodes.

In contrast, TD(0) and normalized ADP also reach the target region in every episode but require more episodes before producing approximately repeatable trajectories, as shown in Figs. 5 and 6. Although both comparative methods eventually generate stabilizing trajectories, their critic weights continue to exhibit small but persistent episode-to-episode adaptations. This suggests that reducing the sample-wise Bellman residual alone does not necessarily lead to consistent critic convergence

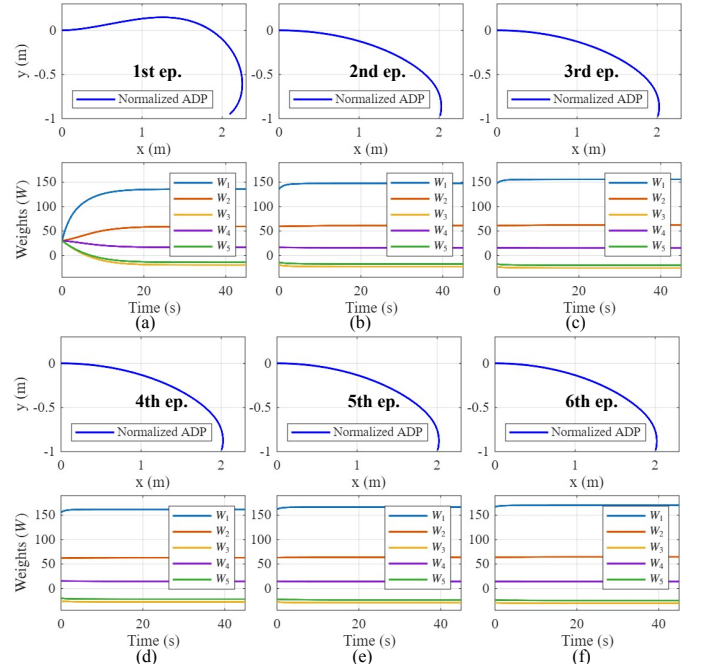


Fig. 6: Learning behavior of normalized ADP across repeated episodes: (a) 1st, (b) 2nd, (c) 3rd, (d) 4th, (e) 5th, and (f) 6th episodes. Each panel shows the AMR trajectory and critic-weight evolution during the corresponding episode.

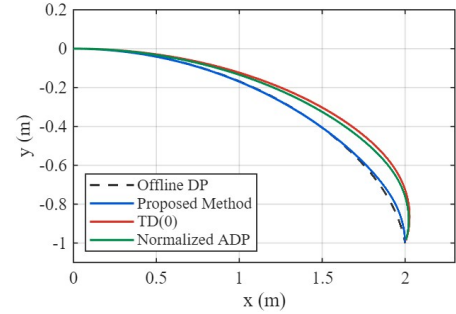


Fig. 7: Trajectory comparison between the offline DP reference and the learned policies.

across repeated task executions.

Following learning across repeated episodes, the learned policies were compared with the offline DP reference solution. For each method, the policy obtained from its final reported episode was used: the 3rd episode for the proposed method and the 6th episode for TD(0) and normalized ADP. These episodes correspond to the points at which the respective methods exhibited approximately repeatable closed-loop behavior. Figures 7 and 8 show the trajectory and control input comparisons, respectively.

Figure 7 shows that all methods reach the target region, while the trajectory generated by the proposed method remains closer to the offline DP reference. Figure 8 further shows that the proposed method generates linear- and angular-velocity profiles that more closely match those produced by the DP

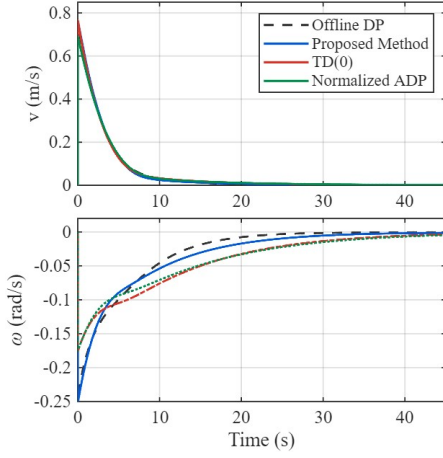


Fig. 8: Control input comparison between the offline DP reference and the learned policies.

reference policy.

The quantitative comparison is summarized in Table II. The trajectory RMSE is computed from the position deviation relative to the DP trajectory, and the control input RMSE is computed from the differences in linear and angular velocities relative to the DP reference.

As reported in Table II, the proposed method achieves the lowest trajectory and control input RMSEs among the compared methods. Its trajectory RMSE is approximately 73.8% and 74.9% lower than those of TD(0) and normalized ADP, respectively. Its control input RMSE is also approximately 59.9% and 60.6% lower than those of TD(0) and normalized ADP, respectively. Overall, the results demonstrate more repeatable behavior across episodes and closer agreement with the offline DP reference for the tested AMR target-reaching task.

2) *Gain Sensitivity Analysis*: To support the nominal gain selection used in the experiments, a simulation-based gain sensitivity analysis was conducted using the nominal learning gains specified above. For each method, the corresponding nominal gain was used as the center value and scaled from $0.1\times$ to $10\times$ using 21 logarithmically spaced values. For the proposed method, two cases were considered: varying α_W with fixed μ , and varying μ with fixed α_W . For TD(0) learning and normalized ADP, the critic learning rate α_W was varied.

Figure 9 shows the trajectory RMSE maps over the gain scale and episode index for each method. A gain scale of 1 corresponds to the nominal gain used in the experiments. The red markers identify the gain-scale and episode combination yielding the minimum trajectory RMSE among the evaluated

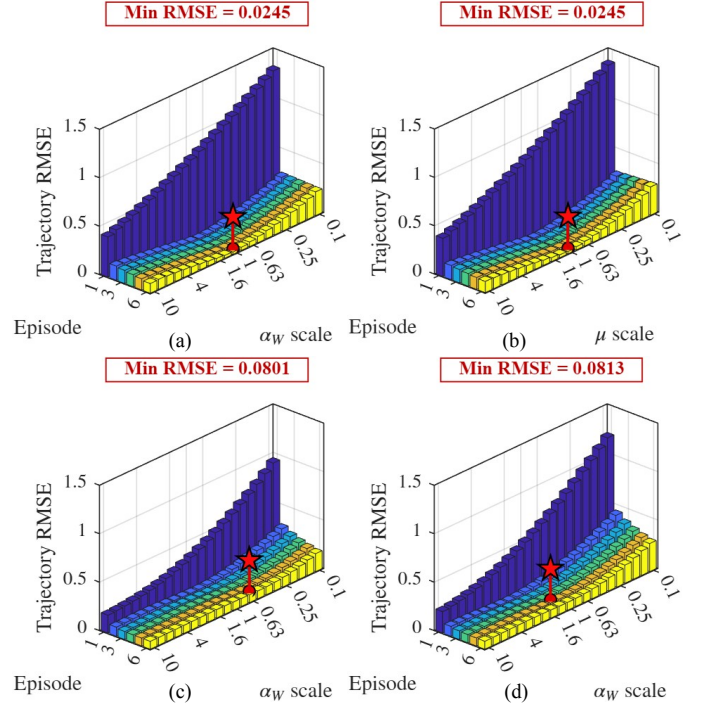


Fig. 9: Gain selection analysis based on trajectory RMSE: (a) proposed α_W , (b) proposed μ , (c) TD(0) learning, and (d) normalized ADP. A gain scale of 1 corresponds to the nominal experimental gain. Red dots indicate the minimum RMSE over the evaluated gain scales and episodes.

cases, rather than indicating the nominal gain. The nominal gains lie within reasonably well-performing regions for their respective methods. Because the same gain-scaling and evaluation procedure was applied to all methods, this analysis reduces the possibility that the experimental comparison was influenced by unfavorable gain settings for the comparative methods.

C. Simulation Analysis

The experimental results demonstrate the closed-loop performance of the learned policies on the AMR platform. However, target-reaching performance alone does not fully explain whether the learned critic approximates the Bellman optimality structure. Therefore, additional simulation analyses are conducted to examine the learned critic from the viewpoint of the Bellman residual and Bellman optimality residual.

1) *Residual-Based Analysis of Experimentally Learned Critics*: The following residual surfaces are generated using the final critic weights obtained from the AMR experiments, with the corresponding experimentally recorded trajectories overlaid. The critic weights are held fixed during evaluation, and the error dynamics are used to compute the one-step transitions over the state-error grid.

First, the Bellman residual along the closed-loop trajectory generated by the learned policy is evaluated. For each sampled

TABLE II: Experimental RMSE comparison with the offline DP reference.

Method	Trajectory RMSE	Control input RMSE
Proposed Method	0.02489	0.00865
TD(0) Learning	0.09489	0.02159
Normalized ADP	0.09924	0.02194

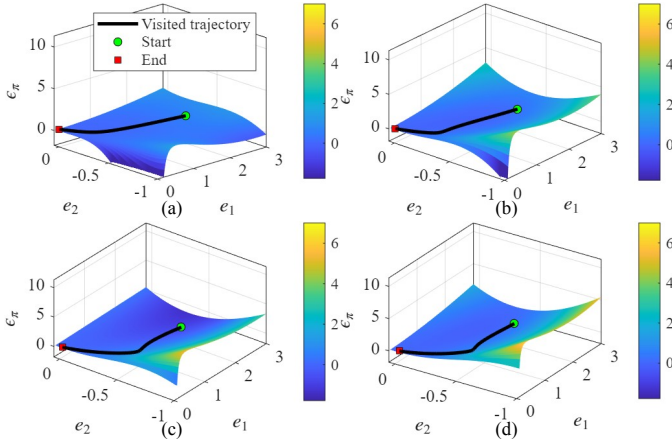


Fig. 10: Bellman residual surfaces ($\varepsilon_\pi(e)$) with the corresponding visited trajectories overlaid: (a) offline DP, (b) proposed method, (c) TD(0) learning, and (d) normalized ADP.

transition $(e_k, e_{k+1}, r(e_k, u_k))$, the policy-dependent Bellman residual is computed as

$$\varepsilon_\pi(e_k) = r(e_k, u_k) + \gamma \hat{f}(e_{k+1}, W) - \hat{f}(e_k, W). \quad (63)$$

Here, u_k denotes the control input generated by the learned policy and applied to the AMR. The subscript π indicates that the residual is evaluated using the closed-loop transition generated by the learned policy. Thus, $\varepsilon_\pi(e_k)$ evaluates whether the learned critic is consistent with the transition resulting from the applied control input.

Figure 10 shows the Bellman residual surfaces obtained after learning from the original trajectory. Around the visited trajectory region, the proposed method and the comparative methods all exhibit relatively small Bellman residuals. This result indicates that each method can reduce the sample-wise Bellman residual along the learned closed-loop trajectory. However, a small value of $\varepsilon_\pi(e_k)$ only indicates consistency with the transition generated by the applied input u_k and does not verify whether that input minimizes the Bellman backup.

To examine Bellman optimality consistency independently of the control input generated by the learned policy, the Bellman optimality residual is evaluated as

$$\varepsilon^*(e_k) = \min_{u \in \mathcal{U}_g} \{r(e_k, u) + \gamma \hat{f}(e_{k+1}(u), W)\} - \hat{f}(e_k, W), \quad (64)$$

where $\mathcal{U}_g$ denotes the finite input grid constructed by discretizing the admissible ranges of the linear and angular velocities, and $e_{k+1}(u)$ denotes the one-step successor error state corresponding to the candidate input u .

At each error-state grid point, the one-step Bellman backup is evaluated for every input candidate in $\mathcal{U}_g$, and its minimum value is selected. Thus, the minimization in (64) represents an exhaustive search over the discretized input space and provides a numerical approximation of the minimization in the Bellman optimality equation.

Figure 11 compares the Bellman optimality residual surfaces. Although the Bellman residuals in Fig. 10 are small near

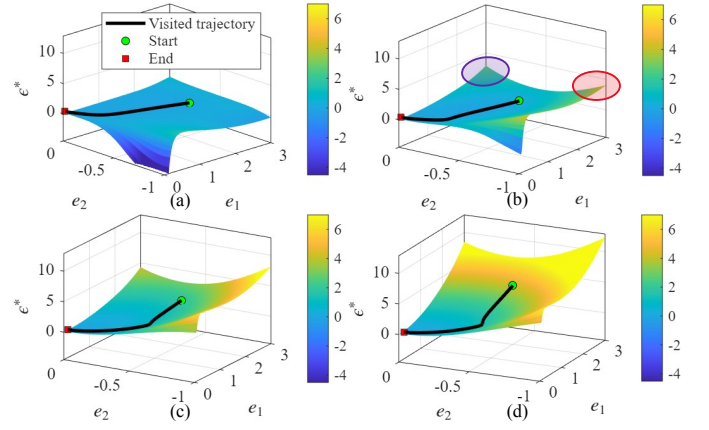


Fig. 11: Bellman optimality residual surfaces ($\varepsilon^*(e)$) with the corresponding visited trajectories overlaid: (a) offline DP, (b) proposed method, (c) TD(0) learning, and (d) normalized ADP. The highlighted regions in (b) are used for the critic-freeze and additional-learning tests.

the visited trajectories, the optimality residual surfaces reveal clearer differences among the learned critics. The proposed method produces a smoother residual surface that is more consistent with the offline DP reference, whereas TD(0) learning and normalized ADP exhibit larger residual variations. This indicates that reducing the Bellman residual along sampled transitions does not necessarily imply that the learned critic approximates the Bellman optimality structure.

The highlighted regions in Fig. 11(b) are used for the subsequent simulation tests. The red region corresponds to a region with relatively small $|\varepsilon^*|$ and is selected for the critic-freeze test described in Section V-C2, whereas the purple region corresponds to a region with relatively large $|\varepsilon^*|$ and is selected for the additional-learning test described in Section V-C3.

2) Critic-Freeze Test: To examine whether the learned critic provides useful policy information for states not encountered during the original learning process, a critic-freeze test is conducted. In this test, the critic weights obtained from the original learning trajectory are held fixed, and the resulting policy is evaluated, without further adaptation, from a new initial error condition outside the originally visited trajectory:

$$e_0 = \begin{bmatrix} e_{1,0} \\ e_{2,0} \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}. \quad (65)$$

Figure 12 shows the resulting trajectories. The trajectory generated by the proposed method remains close to the offline DP trajectory, while the comparative methods show larger deviations. Since the critic is frozen during this test, the result indicates that the critic learned by the proposed method provides a more useful policy representation in the selected off-trajectory region. This supports the interpretation that the proposed method improves Bellman optimality consistency, rather than only reducing the sample-wise Bellman residual along the original trajectory.

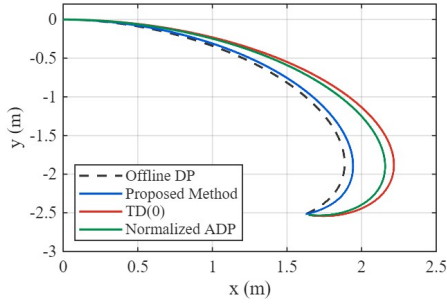


Fig. 12: Closed-loop trajectory comparison for the critic-free test from the new initial error condition $e_0 = [3, -1]^T$.

3) Additional Learning in a Previously Unvisited Region:

The previous analysis shows that the learned critics can exhibit different Bellman optimality residual characteristics even when their Bellman residuals are small near the originally visited trajectory. To further evaluate the online learning behavior of the three methods in a previously unvisited trajectory region, additional learning is conducted from the following initial error condition:

$$e_0 = \begin{bmatrix} e_{1,0} \\ e_{2,0} \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}. \quad (66)$$

This initial condition lies in an error-state region that was not visited during the original learning process and in which all three learned critics initially exhibit relatively large Bellman optimality residuals.

Figure 13 compares the offline DP reference surface with the Bellman optimality residual surfaces obtained after additional learning. The proposed method is evaluated after the 4th additional-learning episode, whereas TD(0) and normalized ADP are evaluated after the 6th episode. Near the newly visited trajectory, the proposed method produces a more pronounced reduction in the Bellman optimality residual magnitude, while the comparative methods exhibit smaller changes despite being evaluated after additional episodes.

Figure 14 shows the evolution of the Bellman optimality residual surface across successive additional-learning episodes of the proposed method. The residual magnitude progressively decreases around the newly visited trajectory without a substantial increase around the original learning trajectory. This indicates that the proposed method can improve local Bellman optimality consistency in a newly encountered trajectory region while largely preserving the consistency acquired in the previously visited region.

To quantify these observations, the absolute Bellman optimality residual is accumulated over the trajectory-associated grid $\mathcal{G}_{\text{traj}}$ and the entire evaluation grid $\mathcal{G}_{\Omega}$:

$$I_{|\varepsilon^*|, \text{traj}} = A \sum_{e_{ij} \in \mathcal{G}_{\text{traj}}} |\varepsilon^*(e_{ij})|, \quad (67)$$

$$I_{|\varepsilon^*|, \Omega} = A \sum_{e_{ij} \in \mathcal{G}_{\Omega}} |\varepsilon^*(e_{ij})|, \quad (68)$$

where A denotes the area of each cell in the uniform e_1 - e_2 evaluation grid. The region $e_1 < 0.1$ is excluded from both

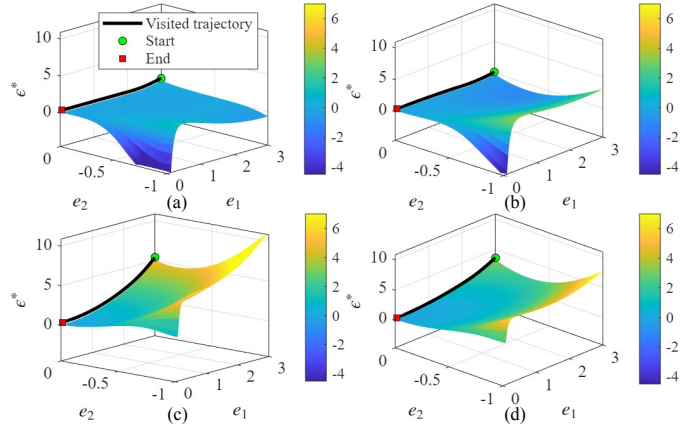


Fig. 13: Bellman optimality residual surfaces ($\varepsilon^*(e)$) for the offline DP reference and the learned critics after additional learning from $e_0 = [3, 0]^T$: (a) offline DP, (b) proposed method after the 4th additional-learning episode, (c) TD(0) learning after the 6th episode, and (d) normalized ADP after the 6th episode.

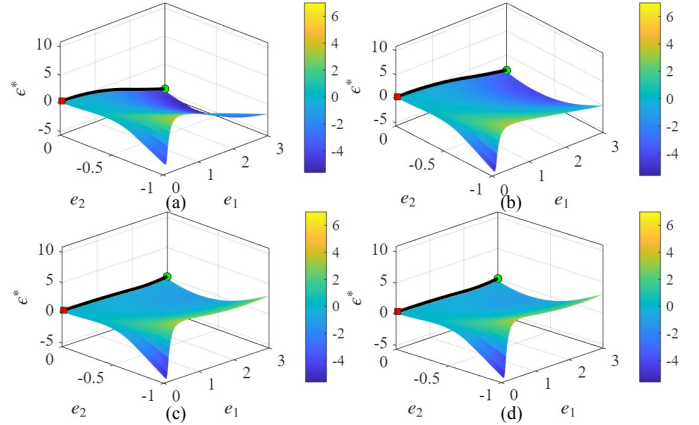


Fig. 14: Evolution of the Bellman optimality residual surface ($\varepsilon^*(e)$) across additional-learning episodes of the proposed method from $e_0 = [3, 0]^T$: (a) after the 1st, (b) 2nd, (c) 3rd, and (d) 4th episodes.

measures to avoid numerical sensitivity near the coordinate singularity at $e_1 = 0$, where the polar error dynamics contain inverse-distance terms. The first measure evaluates the corresponding trajectory-associated region, whereas the second compares all critics over a common evaluation region.

Figure 15 shows that the proposed method achieves the lowest accumulated residual among the three learning methods for both the trajectory-associated and full-grid evaluations, before and after additional learning. Before additional learning, its trajectory-associated accumulated residual is 83.1% and 84.4% lower than those of TD(0) and normalized ADP, respectively, while its full-grid accumulated residual is 52.2% and 62.4% lower. Over the common evaluation grid, additional learning reduces $I_{|\varepsilon^*|, \Omega}$ by 8.8% for the proposed method and 27.2% for normalized ADP, while increasing it by 22.5%

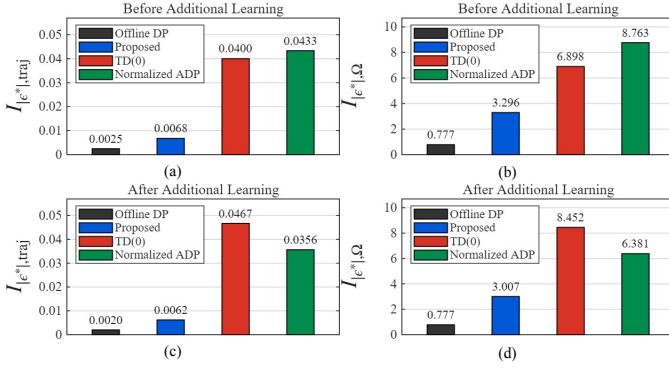


Fig. 15: Accumulated absolute Bellman optimality residuals: (a) trajectory-associated and (b) full-grid measures before additional learning; (c) trajectory-associated and (d) full-grid measures after additional learning.

for TD(0). Although normalized ADP shows a larger relative reduction, its final accumulated residual remains higher than that of the proposed method.

The increase for TD(0) is consistent with the larger residual magnitudes observed away from the additional-learning trajectory in Figs. 11(c) and 13(c). This indicates that adaptation along a new trajectory can degrade Bellman optimality consistency elsewhere. In contrast, the proposed method improves consistency near the newly visited trajectory while also reducing the accumulated residual over the entire evaluation grid.

Overall, the simulation results indicate that the magnitude of the Bellman residual alone does not fully characterize the quality of a learned critic for control. Compared with TD(0) and normalized ADP, the residual-surface and critic-freeze tests show that the proposed critic is more consistent with the Bellman optimality structure and provides policy behavior more closely aligned with the offline DP reference beyond the original learning trajectory. The additional-learning test further shows that this local consistency can be extended to a newly visited trajectory region without substantially degrading that in the original region.

VI. CONCLUSIONS

This paper presented a constrained online critic-learning method for discrete-time control-affine nonlinear systems, using the one-step Bellman backup as the optimization objective and value consistency as an equality constraint. The resulting learning law updates the critic weights and Bellman multiplier from sampled transitions and supports policy improvement through the learned critic gradient. Under the stated assumptions, Lyapunov analysis established uniform ultimate boundedness of the critic-weight and multiplier errors.

Real-world AMR experiments demonstrated closer agreement with the offline DP reference, with trajectory RMSE reductions of 73.8% and 74.9% relative to TD(0) and normalized ADP, respectively. Complementary simulations showed that the accumulated Bellman optimality residual over the

evaluated state-error grid was 52.2% and 62.4% lower than those of TD(0) and normalized ADP, respectively. Residual-surface and critic-freeze analyses further showed that the learned critic provided policy behavior more closely aligned with the offline DP reference beyond the original learning trajectory. Additional learning improved Bellman optimality consistency in a newly visited region without substantially degrading that achieved near the original trajectory.

These results support explicitly incorporating the Bellman optimality structure into online critic learning. While the residual-surface and critic-freeze results demonstrate that the improved Bellman optimality consistency extends beyond the sampled learning trajectory within the evaluated grid, establishing such consistency globally over the entire state space remains beyond the scope of this study. In addition, the current validation focuses on a low-dimensional AMR error system. Accordingly, future work will develop a joint analysis of learning and closed-loop stability under verifiable excitation and update conditions, extend the method to higher-dimensional systems using sampling-based residual evaluation, and investigate continual learning over multiple trajectories while preserving consistency in previously visited regions.

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